

NPHI101: Engineering Physics
End-Semester Examination 2025-26 (Monsoon Semester)

Total Marks: 50

Time: 2 hours

Use separate answer sheets for different parts. Mention the Part No. at the top.
All the symbols have their usual meaning.

PART 1 (CLASSICAL MECHANICS AND ELECTRODYNAMICS)

1. A long cylinder carries a charge density that is proportional to the distance from the axis, $\rho = ks$, for some constant k ; s is the radial distance from the axis. Find the electric field inside this cylinder.

Marks: 5

$$\frac{k s^2}{3 \epsilon_0} \hat{s}$$

2. (a) Write down the continuity equation for charge conservation. (b) Derive Maxwell's correction for the equation for $\vec{\nabla} \times \vec{B}$, starting from the continuity equation and using the example of a capacitor; $\vec{B}$ is the magnetic field. Write down the final form of the equation for $\vec{\nabla} \times \vec{B}$. Marks: 2+3+1=6

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 (\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t})$$

3. Which one of these is an impossible electrostatic field? (a) $\vec{E} = k[xy\hat{x} + 2yz\hat{y} + 3xz\hat{z}]$; (b) $\vec{E} = k[y^2\hat{x} + (2xy + z^2)\hat{y} + 2yz\hat{z}]$. Here, k is a constant with appropriate units. Prove your answer.

Marks: 1+4=5

$$\vec{\nabla} \times \vec{E} = 0$$

PART 2 (THERMAL AND STATISTICAL PHYSICS)

1. The partition function of a canonical ensemble is given by $Z = \exp(bT^3V^2)$. Here, b is a constant, T is the absolute temperature, and V is the volume. Find Helmholtz free energy, internal energy, and entropy. $F = -bk_B T^3 V^2$ $U = \frac{3bV^2}{k_B T^4}$ $S = \frac{1}{T}(-F + U)$ Marks: 1.5+2+1.5=5

2. The single-particle grand partition function is given by $Z_i = \sum_{n_i} \Omega(\epsilon_i, n_i) e^{-\beta n_i(\epsilon_i - \mu)}$. Using this, find the average occupation number of bosons in a state with energy ϵ . Marks: 3

$$\frac{e^{\beta(\epsilon - \mu)}}{e^{\beta(\epsilon - \mu)} - 1}$$

3. (a) A system of non-interacting spin $\frac{1}{2}$ particles is in thermal equilibrium. The single-particle energy level $\epsilon = 0.15$ eV has a fractional occupation $\langle n \rangle = 0.35$. The chemical potential is $\mu = 0.10$ eV. Using the appropriate statistics, find the temperature. ($k_B = 8.62 \times 10^{-5}$ eV/K) 936.9353

- (b) Graphically show the distribution at this temperature with appropriate labeling. Marks: 3+2=5

4. For an open thermodynamic system, the first law is $dU = TdS - PdV + \mu dN$. Using this relation, derive the expression for the differential of Gibbs free energy (dG) for an isobaric process. Marks: 2

$$dG = \mu dN - S dT$$

5. Show that Planck's law of black body radiation reduces to Wien's law in the short-wavelength limit.

Marks: 2

$$dG = U - TS + PV$$

PART 3 (MODERN PHYSICS)

1. Consider a quantum particle of mass m within an infinite square well potential (in 1-D), which is in the first excited state. Write down the normalized wave function. Calculate the expectation value of the energy using its definition. $\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$ $E = \frac{4\pi^2 \hbar^2}{2ma^2}$ Marks: 2+2=4

2. Demonstrate the Hall effect in the free electron model and derive the expression of the Hall constant using a schematic diagram and mathematical calculations. $R_{H, \text{Boson's symmetric}}$ Marks: 6

3. Consider $\psi(x) = Ax^2$, where A is constant. What kind of quantum particle can it represent? Marks: 2

4. What is the full form of LASER? What are the three essential components of a laser? Marks: 2+3=5

Light Amplification by Stimulated
emission of radiation

1

laser med
pumping

optical med